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Abstract — Micro aerial vehicles are becoming increasingly important in search and rescue operations due to their agility, speed and ability to access confined spaces or hazardous areas. However, designing lightweight

aerial systems presents significant structural, aerodynamic and computational challenges. This work addresses two key limitations in many low-cost aerial systems under two kilogrammes: their lack of structural durability during flight through rough terrains and inability to replan paths dynamically when new victims or obstacles are detected. We present a fully customised drone built from scratch using only commonly available components and materials, emphasising modularity, low cost and ease of assembly. The structural frame is reinforced with lightweight yet durable materials to withstand impact, while the onboard control system is powered entirely by free, open-source software solutions. The proposed system demonstrates real-time perception and adaptive navigation capabilities without relying on expensive hardware accelerators by offering an affordable and practical solution for real-world search and rescue missions.

Keywords (separated by '-') Autonomous UAV - Path planning - Structural analysis

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Design and Structural Validation of a Micro-UAV with Onboard Dynamic Route Planning

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Inbazhagan Ravikumar, Ram Sundhar, and Narendhiran Vijayakumar

AQ1

Abstract Micro aerial vehicles are becoming increasingly important in search and rescue operations due to their agility, speed and ability to access confined spaces or hazardous areas. However, designing lightweight aerial systems presents significant structural, aerodynamic and computational challenges. This work addresses two key limitations in many low-cost aerial systems under two kilogrammes: their lack of structural durability during flight through rough terrains and inability to replan paths dynamically when new victims or obstacles are detected. We present a fully customised drone built from scratch using only commonly available components and materials, emphasising modularity, low cost and ease of assembly. The structural frame is reinforced with lightweight yet durable materials to withstand impact, while the onboard control system is powered entirely by free, open-source software solutions. The proposed system demonstrates real-time perception and adaptive navigation capabilities without relying on expensive hardware accelerators by offering an affordable and practical solution for real-world search and rescue missions.

Keywords Autonomous UAV · Path planning · Structural analysis

1 Introduction

Micro-class quadcopters ($\leq$2 kg) are often used in search and rescue (SAR) missions due to their agility and rapid deployment [1]. However, they face structural and navigational challenges, especially in rugged terrain. Lightweight designs often compromise stiffness and impact resistance, increasing failure risks during hard landings.

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A structurally efficient quadcopter frame was developed, combining lightweight materials with improved impact resistance and aerodynamic shaping. The design

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was validated through structural and aerodynamic analyses to ensure reliability in demanding conditions [2].

Object detection models for different UAV applications predominantly involve trade-offs between one-stage (e.g. YOLO variants) and two-stage (e.g. Faster R-CNN) detectors. Lightweight CNNs such as SlimYOLOv3 enable efficient real-time victim detection on embedded platforms [3]. Combined with dynamic path replanning algorithms, ranging from greedy nearest-neighbour heuristics to optimal travelling-salesman solvers, drones can swiftly update their routes based on live detections, markedly enhancing mission responsiveness [4].

This paper proposes an integrated quadcopter solution combining robust structural design with optimal onboard vision-driven autonomy, validated through structural analyses and experimental SAR scenario testing.

2 Related Work

This section explores prior research and technologies relevant to developing autonomous search and rescue (SAR) drones by examining existing work in three key areas: structural resilience and rigidity in UAV design, onboard vision systems and component selection and dynamic path planning algorithms.

2.1 Structural Resilience and Rigidity

Drone frame design now emphasises structural resilience using lightweight composites like carbon fibre for their high strength-to-weight ratio [5]. Features like flexible landing gear and hybrid polymer joints improve shock absorption during rough landings. Finite Element Analysis (FEA) is widely used to evaluate stress and vibration behaviour, ensuring reliability across various conditions. However, balancing rigidity, weight reduction and ease of manufacturing remain a key design challenge [6].

2.2 Onboard Vision and System Selection

Computer vision models for UAVs predominantly involve trade-offs between one-stage (e.g. YOLO) and two-stage (e.g. Faster R-CNN) detectors. Two-stage algorithms generally deliver higher accuracy but impose prohibitive computational loads on embedded UAV hardware [7]. By contrast, one-stage detectors achieve a more suitable speed–accuracy balance for real-time flight. Benchmark studies on platforms such as NVIDIA Jetson and Raspberry Pi series reveal marked differences in inference throughput and power efficiency, guiding optimal subsystem selection for

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onboard deployment [8]. Lightweight communication protocols like MAVLink efficiently handle telemetry and command exchange between UAV and ground station under strict energy constraints [9].

2.3 Dynamic Path Planning

Dynamic path planners for search and rescue UAVs range from ultra-light greedy schemes to heavy optimisation. Nearest-neighbour routes refined only by minor post-processing enable millisecond replans but inflate tour length by 15% as target density rises [10]. Mixed-Integer Linear Programming (MILP)-based coverage formulations like the CUAVRP algorithm [11] minimise total distance yet become intractable once waypoint counts climb past a few dozen, breaching real-time limits. Meta-heuristic multi-objective strategies cut cost but still demand seconds to minutes of CPU time on embedded hardware [12]. This raises the concern about balancing the need for computational efficiency in path planning algorithms.

3 Methodology

This section details the methodology for designing and developing autonomous search and rescue (SAR) drones. We cover structural design, onboard vision and system selection and dynamic path planning.

3.1 Structural Resilience and Rigidity

The main objective of the structural design was to create a lightweight, rigid and durable frame capable of withstanding flight loads, landing impacts and collisions.

A quadrotor with an X-frame configuration was chosen for stability and sleeker design. The structure used a cross-frame layout with a double-layer sandwich hub, allowing compact component placement and high bending stiffness. Four-leg landing gear was designed to withstand drops from height and enable landing on uneven and slant terrain, as per the operation to be carried out.

Carbon-fibre composites were used for the hub and extrusion arms due to their high strength-to-weight ratio. TPU was selected for 3D-printed landing gear shoes with gyroid infill pattern for impact absorption. A vacuum-formed PETG dome protects the electronics and reduces drag. Various structural analyses were carried out to optimise the weight and strength of each mechanical component, and computational fluid dynamics was carried out to study fluid-body interactions.

The airframe was tested under simulated flight loads (40 N thrust, 20 N weight) to verify structural integrity and ensure an adequate safety factor under normal operating

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conditions. LS-Dyna simulations assessed landing gear durability under 15m drop tests. Test conditions were iteratively refined to improve energy absorption using foam-encased rods [13]. Drone arms were analysed for natural frequencies and mode shapes to avoid resonance. A motor speed of 1000 rpm (166.67 Hz) was used as the baseline for vibrational safety [14].

CFD analysis was performed using mesh motion to study propeller clearance in hover. Additional flow visualisation over the UAV body was done to understand cruise-phase aerodynamics. No closed-form parametric equations were used. Instead, design parameters were swept on a manually generated grid, and each configuration was analysed in ANSYS under the specified loads and constraints.

3.2 Onboard Vision and Subsystem Architecture

Figure 1 shows the avionics stack for cameras, compute boards, RF links and power modules. Perception runs on a Raspberry Pi 5. A Pixhawk 6C handles flight control. They exchange data over a 921 kbit/s serial link using ROS 2 XRCE-DDS. A micro-agent on the Pi publishes the/roi_array detections topic to PX4, where the uXRCE-DDS client converts it into the uORB vehicle_trajectory_waypoint. The bridge bypasses MAVLink parsing for real-time ROI waypoint ingestion. Telemetry topics such as vehicle_gps_position and battery_status stream back to QGroundControl for live path visualisation in Fig. 2.

Onboard inference performance of SSDMobileNetV3-FPN Lite and YOLOv8n was benchmarked using a custom aerial dataset acquired during mission flights. Converting both networks to the NCNN runtime cuts latency by roughly 5$\times$ without significant loss in precision; the resulting YOLOv8n in NCNN format runs at about 20 FPS full-resolution and publishes geo-tagged detections to the planner. Fail-safes are embedded in PX4: a GPS bounding-box watchdog triggers Return-to-Home on geofence breach, loss of RF link escalates from hover to RTH after 10 s and battery thresholds at 20% (warning) and 15% (auto-land) protect endurance. This vision pipeline, communication bus and safety logic deliver a fully self-contained SAR-grade autonomous UAV.

3.3 Dynamic Path Planning

The proposed dynamic path planning methodology enables the UAV to navigate changing environments efficiently. The algorithm alternates between SWEEP and SERVICE. It first overlays a lawn-mower grid on the geofence; flying this grid alone guarantees full coverage. If no targets are seen, the drone remains in SWEEP and proceeds to the next grid waypoint. Once a ROI appears, the mode switches to

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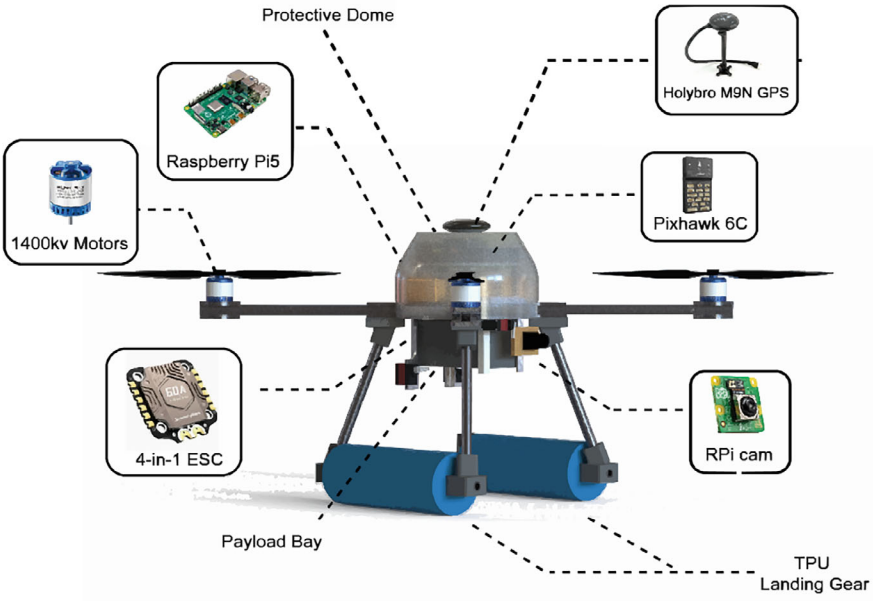


Fig. 1 Visual abstract of the custom SAR quadcopter and its core hardware components

SERVICE: the three nearest ROIs are selected by a quick nearest-neighbour test, visited in that order and removed from the queue. Newly detected ROIs are inserted immediately, preventing stale routes. When the list is empty, the vehicle returns to the grid.

Algorithm 1 Light-Weight Incremental Greedy (LIG) re-routing algorithm

```
LM ← LawnMowerGrid(P, Δ)                    // pre-compute lawn-mower path
S ← ∅; i ← 0; mode ← SWEEP
while the mission is active do
    S ← S ∪ NewDetections()                  // merge novel ROI centroids
    mode ← SWEEP if S = ∅ else SERVICE
    if mode = SWEEP then
        FlyTo(LM[i])
        if Reached(LM[i]) then i ← i + 1
    else                                     // SERVICE mode
        B ← K nearest points in S to Q       // pure NN scan
        for each b in B (NN order) do
            FlyTo(b)
            S ← S \{b}
            S ← S ∪ NewDetections()
            if S = ∅ then break
        end for
    end if
    ClipPathToFence( P )
end while
```

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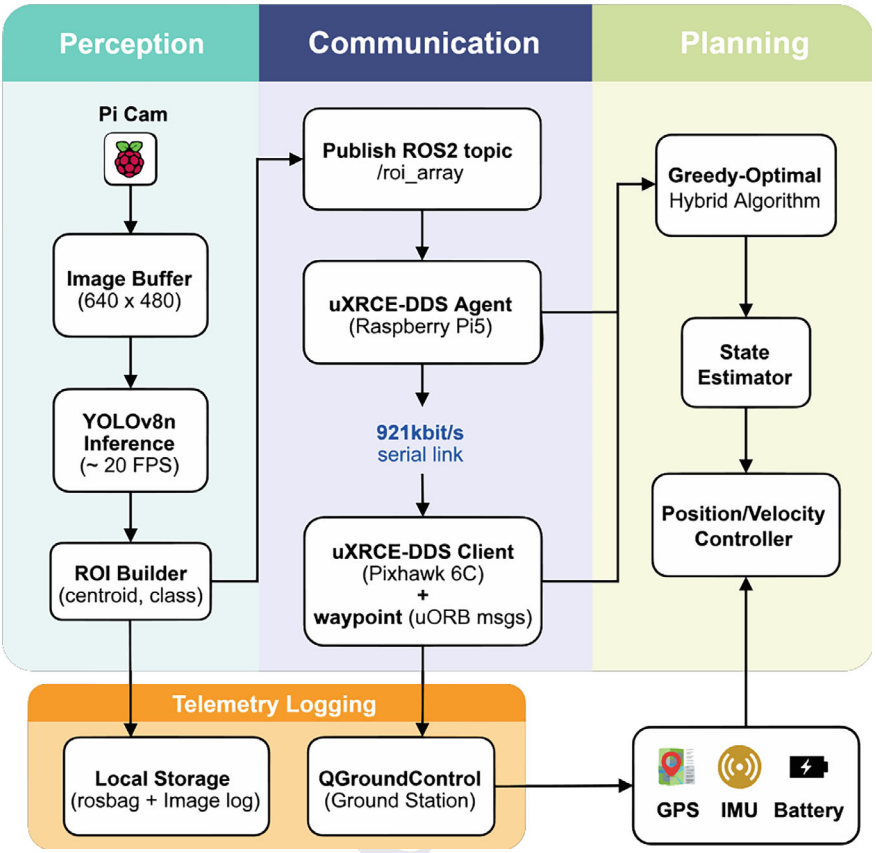


Fig. 2 Overview of the onboard autonomous navigation pipeline

4 Results and Discussion

The development of an autonomous SAR drone presents a multifaceted engineering challenge. The proposed methodology balances performance, cost and complexity to achieve a functional prototype. Under the given loading conditions, the structure showed a maximum displacement of 1 mm at the motor arm ends and a peak stress of 0.4 MPa. The computed factor of safety was approximately 3.5, confirming compliance with UAV design requirements.

Simulated drop tests confirmed that the landing gear could withstand impacts from 15 m. Encasing longitudinal rods with foam significantly enhanced damping and reduced impact stresses, improving resilience on rough terrain, as shown in Fig. 3.

The modal analysis confirmed that all the arms' primary vibrational and torsional modes were well above the motor operating frequency of 166.67 Hz, thus eliminating

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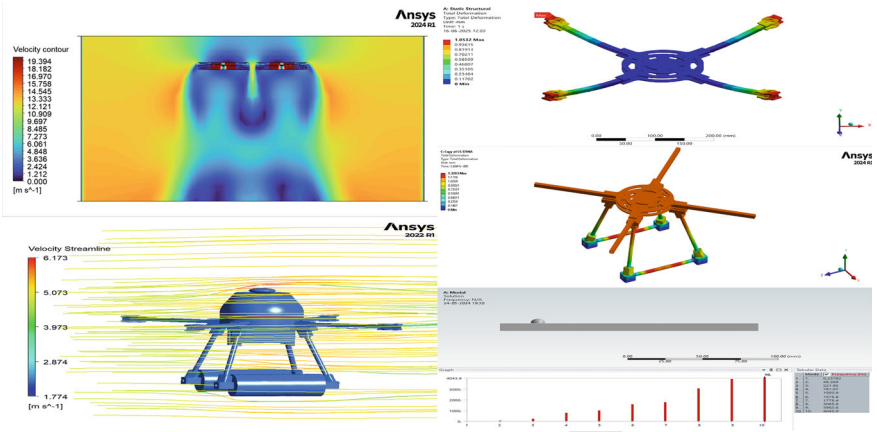


Fig. 3 Velocity contour (top left) and streamline (bottom left) over UAV; airframe deformation (top right), 15 m impact response (mid right) and rotor arm frequency modes (bottom right)

the risk of resonance during flight, as shown in Table 1. Mesh motion study results indicated a cumulative thrust of 56 N (as shown in Table 2), validating a thrust–weight ratio 2. The propeller moment was 0.0083 Nm, confirming torque balance. Further aerodynamic refinements reduced the drag coefficient from 0.7 to 0.59, improving flight efficiency.

Table 3 shows that converting YOLOv8n to NCNN yields a 34% mAP gain over SSD-MobileNetV3-FPN Lite while adding only 16 ms latency. With 8.7 GFLOPs and ~14% more parameters still within Raspberry Pi 5 limits, we use YOLOv8n over SSD-MobileNetV3. As shown in Table 4, the nearest-neighbour heuristic runs in ~2–3 ms but produces the longest tours, whereas Concorde gives the shortest tour yet

Table 1 Modes of frequency obtained from harmonic analysis

Mode	Frequency (Hz)	Deformation axis	Maximum deformation (in mm)
1	299.67	lateral axis	277.67
2	316.31	vertical axis	281.07
3	1830.60	lateral axis	279.12

Table 2 CFD results for propeller analysis (mesh motion study)

Body	Thrust (in N)
Propeller 1	13.992181
Propeller 2	13.970296
Propeller 3	13.970296
Propeller 4	14.007305

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Table 3 Performance comparison of single-shot detectors on Raspberry Pi 5

Model	mAP	GFLOPs	Size (M)	Latency*
SSD-MobileNetV3-FPN Lite	0.607	1.1	2.8	32
YOLOv8n	0.734	8.7	3.2	48

* Single-thread inference (ms) on Raspberry Pi 5, 640 × 480 input, averaged over 300 frames after NCNN conversion

Table 4 Onboard planning and the time taken by different route-planning algorithms

Algorithm	Approach	Time* (in sec)	Mean onboard latency (in ms)
Nearest-neighbour heuristic (NNH)	Purely greedy	415	2
Concorde MILP for TSP	Globally optimal	342	70
LIG (proposed)	Hybrid	355	4

* Simulated time to visit 25 ROI waypoints in a 600 m × 400 m search box at 5 m s^{-1}

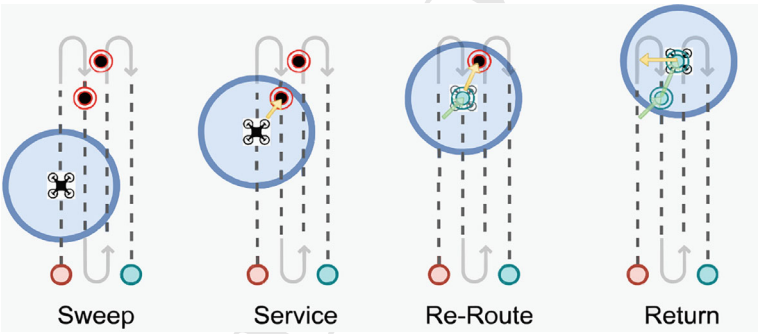


Fig. 4 Re-routing sequence of the proposed planner

requires ~70 ms per update onboard [15]. Figure 4 shows re-routing by the proposed algorithm in <5 ms while closing ~85% of the gap to the optimal tour length.

5 Conclusion

This work presents a sub-2 kg quadrotor that addresses three bottlenecks: fragile airframe, slow onboard perception and rigid route planning. Carbon-fibre arms and TPU skids enhance crash tolerance, and FEA and CFD analysis confirm adequate structural stiffness and aerodynamic performance. Converting YOLOv8n to NCNN improves mAP by 34% over SSD-MobileNetV3-FPN Lite while keeping latency under 50 ms. Detections drive a hybrid greedy-optimal planner that updates tours

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and reduces detour distance. Future work will improve detection under low visibility, enable coordinated multi-UAV operation and refine the design for enhanced portability and complex-path navigation.

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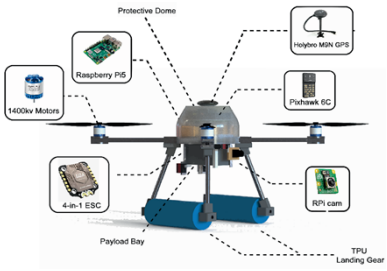
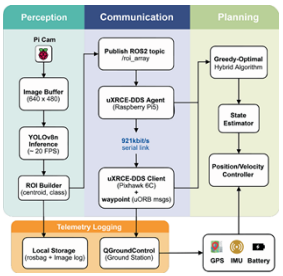
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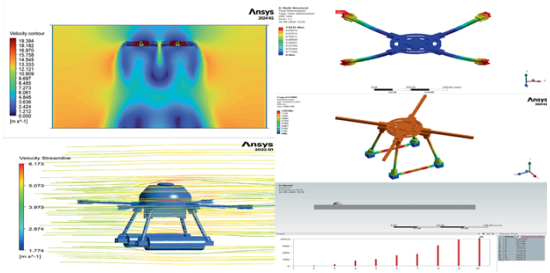
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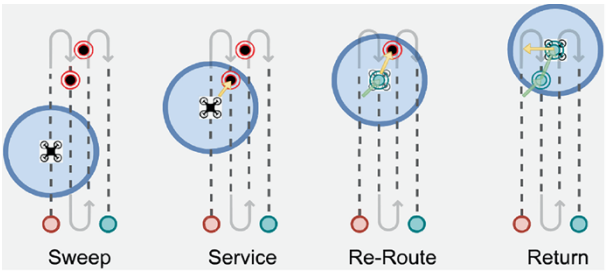
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5	Fig1		This drone includes a protective dome covering its central components. It is equipped with four 1400-kilovolt motors, each connected to a propeller. The drone's flight control system uses a Pixhawk 6C autopilot and a 4-in-1 electronic speed controller rated at 60 amperes. Navigation is supported by a Holybro M9N GPS module. The onboard computer is a Raspberry Pi 5, which interfaces with an RPi camera for visual input. The drone has a payload bay for carrying equipment or cargo and uses TPU material for its landing gear.
6	Fig2		This diagram outlines a system workflow divided into Perception, Communication, Planning, and Telemetry Logging components. In Perception, a Pi Camera captures images at 640 by 480 resolution, processed by YOLOv8n for inference at about 20 frames per second. The Region of Interest Builder extracts centroid and class data from these images. In Communication, the ROI data is published as a ROS2 topic named "roi_array" by a uXRCE-DDS Agent running on a Raspberry Pi 5. This agent sends data over a 921 kilobits per second serial link to a uXRCE-DDS Client on a Pixhawk 6C, which also receives waypoint messages labeled as uORB messages. In Planning, the system uses a Greedy-Optimal Hybrid Algorithm feeding into a State Estimator,

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			which then controls Position and Velocity. Telemetry Logging records data locally as rosbag and image logs, and sends information to QGroundControl at the Ground Station. The Ground Station communicates with GPS, IMU, and Battery modules, which provide input to the Position/Velocity Controller.
5	Figa	**Algorithm 1** Light-Weight Incremental Greedy (LIG) re-routing algorithm **LM ← LawnMowerGrid(P, Δ)** // pre-compute lawn-mower path **S ← ∅; i ← 0; mode ← SWEEP** **while** the mission is active **do** **S ← S ∪ NewDetections()** // merge novel ROI centroids **mode ← SWEEP if S = ∅ else SERVICE** **if mode = SWEEP then** **FlyTo(LM[i])** **if Reached(LM[i]) then i ← i + 1** **else** // SERVICE mode **B ← K nearest points in S to Q** // pure NN scan for each **b** in **B (NN order) do** **FlyTo(b)** **S ← S \{b}** **S ← S ∪ NewDetections()** **if S = ∅ then break** **end for** **end if** **ClipPathToFence(P)** **end while**	The algorithm describes a light-weight incremental greedy re-routing process for a mission involving a lawn-mower path and new detections. It starts by pre-computing a lawn-mower path, then enters a loop while the mission is active. In each iteration, it merges new region of interest centroids into the set S. The mode switches between SWEEP when S is empty and SERVICE otherwise. In SWEEP mode, the system flies to the next point in the lawn-mower path and increments the index upon reaching it. In SERVICE mode, it finds the K nearest points in S to a queue Q, then for each point in nearest neighbor order, it flies to the point, removes it from S, adds new detections, and breaks if no detections remain. After each iteration, it clips the path to a fence defined by P. The loop continues until the mission ends.
7	Fig3		The image presents multiple engineering simulation results related to a drone-like structure. The top left section shows a velocity contour map indicating airflow speed around two rectangular objects, with speeds ranging from zero to about 19.4 meters per second. The bottom left section

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			displays velocity streamlines around a drone model, showing airflow patterns with speeds from approximately 1.8 to 6.2 meters per second. The top right section contains two structural deformation analyses of a four-armed frame, with maximum deformation values indicated, showing how the frame bends under load. The bottom right section includes a modal analysis setup with a simple beam and a graph of modal frequencies, showing vibration modes and their corresponding frequencies. Together, these simulations illustrate aerodynamic behavior, structural deformation, and vibrational characteristics of drone components.
8	Fig4		The image presents four drone operation modes labeled Sweep, Service, Re-Route, and Return. Each mode shows a drone navigating between two points marked with red and teal circles connected by dashed lines and arrows indicating movement paths. In Sweep mode, the drone moves along a direct path between the two points, with a large blue circle around the drone indicating its scanning or sensing area. In Service mode, the drone moves toward a red circle with an orange arrow pointing from the drone to the red circle, suggesting interaction or servicing. In Re-Route mode, the drone moves toward the red circle with an orange arrow pointing from the red circle to the drone, indicating a change in route or direction. In Return mode, the drone moves toward the teal

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			circle with a yellow arrow pointing from the drone to the teal circle, indicating a return to the starting or home point. Each mode visually distinguishes the drone's interaction or movement relative to the two points.